

MULLAN PASS, ID, USA

WMO: 726817

Lat: 47.457N

Lon: 115.645W

Elev: 1837

StdP: 81.11

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 24154

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.9	-15.9	-22.7	0.6	-18.0	-19.6	0.8	-14.0	8.1	-5.9	7.3	-5.3	2.6	120	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	25.2	12.7	23.6	12.2	22.2	11.6	14.0	22.3	13.1	21.4	12.3	20.5	1.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	10.1	16.7	9.4	9.2	15.8	8.4	8.6	14.8	45.1	22.3	42.4	21.5	40.1	20.5	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.6	5.0	DB	-21.2	28.3	4.0	1.8	-24.2	29.6	-26.5	30.6	-28.8	31.6	-31.7	32.9	(4)
(5)			WB	-21.6	15.8	3.9	1.1	-24.5	16.6	-26.8	17.3	-29.0	17.9	-31.8	18.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-4.8	-5.2	-2.4	0.7	5.8	9.8	16.0	15.6	10.3	3.2	-2.3	-5.7	(6)
(7)		DBStd	8.95	4.27	4.75	4.02	4.35	5.01	5.11	4.45	4.35	5.44	4.97	4.65	4.38	(7)
(8)		HDD10.0	2920	458	427	384	280	151	66	8	9	64	217	371	487	(8)
(9)		HDD18.3	5477	716	660	642	528	388	260	97	106	247	468	621	746	(9)
(10)		CDD10.0	536	0	0	0	2	21	59	194	181	72	7	0	0	(10)
(11)		CDD18.3	51	0	0	0	0	0	3	25	19	4	0	0	0	(11)
(12)		CDH23.3	167	0	0	0	0	0	14	86	60	7	0	0	0	(12)
(13)		CDH26.7	15	0	0	0	0	0	2	7	6	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.6	2.7	2.8	2.5	2.3	2.3	2.1	2.1	2.3	2.6	2.8	2.5	(14)
(15)	Precipitation	PrecAvg	960	116	89	115	82	75	76	21	25	42	93	131	125	(15)
(16)		PrecMax	1290	224	172	270	207	139	164	53	118	147	258	325	254	(16)
(17)		PrecMin	767	30	20	56	33	20	34	2	0	10	7	58	9	(17)
(18)		PrecStd	138	51	39	52	41	31	32	14	30	32	54	64	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.1	9.7	16.1	20.9	25.2	27.7	27.3	24.3	17.8	11.1	4.1	(19)
(20)			MCWB	0.2	1.5	3.3	6.6	9.8	13.2	14.4	12.5	11.2	8.6	4.7	0.0	(20)
(21)		2%	DB	3.7	3.7	7.2	12.4	17.8	22.1	25.8	24.9	22.1	14.8	7.9	2.2	(21)
(22)			MCWB	0.2	0.1	1.9	5.2	8.8	12.4	13.4	12.2	10.6	7.2	4.0	0.2	(22)
(23)		5%	DB	1.6	1.8	4.9	9.8	15.7	19.3	23.9	23.3	19.9	12.3	6.0	0.8	(23)
(24)			MCWB	-0.5	-0.4	1.1	4.0	7.7	11.0	12.6	11.8	9.6	6.2	2.9	-0.4	(24)
(25)		10%	DB	0.1	0.3	2.8	7.5	13.2	17.3	22.4	21.8	17.8	10.1	3.7	-0.6	(25)
(26)			MCWB	-1.1	-0.9	0.3	2.8	6.6	10.0	12.1	11.2	9.0	5.1	1.3	-1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.3	4.4	7.5	10.9	14.3	16.0	14.4	13.0	10.0	6.8	2.4	(27)
(28)			MCD B	4.3	4.2	7.1	14.7	17.4	23.0	25.1	22.2	20.2	15.5	8.6	2.8	(28)
(29)		2%	WB	1.0	1.0	2.9	5.7	9.4	12.7	14.6	13.4	11.6	8.2	4.9	0.9	(29)
(30)			MCD B	2.3	2.3	5.4	11.1	16.2	20.3	23.1	21.9	19.6	13.1	7.1	1.3	(30)
(31)		5%	WB	-0.1	0.2	1.9	4.5	8.4	11.6	13.6	12.7	10.5	6.8	3.0	-0.6	(31)
(32)			MCD B	1.1	1.3	4.1	9.2	14.4	18.6	21.9	21.2	18.4	11.0	4.8	0.2	(32)
(33)		10%	WB	-1.1	-0.9	0.6	3.1	7.3	10.4	12.7	11.9	9.6	5.6	1.8	-1.5	(33)
(34)			MCD B	-0.1	0.3	2.6	6.4	12.3	16.5	21.0	20.3	16.8	9.3	3.5	-0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	5.5	6.4	7.2	7.6	8.4	8.3	7.3	5.8	4.3	3.9	(35)
(36)			MCDBR	5.0	5.7	7.8	9.7	9.1	9.8	9.0	9.3	9.0	8.1	6.2	5.0	(36)
(37)			MCWBR	3.3	3.9	4.6	5.0	4.5	5.1	4.9	4.7	4.4	4.5	3.9	3.8	(37)
(38)		5% WB	MCD BR	4.4	4.9	7.0	8.9	8.5	9.4	8.9	8.7	8.5	7.3	5.4	4.4	(38)
(39)			MCWBR	3.6	3.8	4.7	4.8	4.4	5.1	5.1	4.6	4.3	4.4	3.8	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.225	0.230	0.256	0.292	0.303	0.307	0.305	0.321	0.298	0.262	0.251	0.222	(40)	
(41)		taud	2.501	2.495	2.496	2.479	2.516	2.535	2.544	2.493	2.546	2.668	2.600	2.558	(41)	
(42)		Ebn at Noon	907	971	983	966	961	955	952	924	922	922	865	875	(42)	
(43)		Edh at Noon	59	75	89	100	101	100	97	98	84	64	55	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.09	1.95	3.10	4.51	5.53	6.10	7.25	6.02	4.23	2.51	1.23	0.86	(44)	
(45)		RadStd	0.13	0.24	0.33	0.40	0.44	0.55	0.48	0.39	0.45	0.29	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A

Nomenclature: See separate page